

ActInf GuestStream 089.1 ~ "Generic framework for a coherent integration

GuestStream | 1:25:58

Hello and welcome. It's September 30th, 2024. We're in Active Inference Guest Stream 89.1, Generic Framework for a Coherent Integration of Experience and Exposure Rating in Reinsurance with Stefan Werninger and also joined by Roland Borghi. So thank you both for joining. Stefan, to you for a presentation and we'll look forward to that and the discussion.

Okay, thank you, Daniel. Welcome everyone. Let me quickly give an overview of what I'm going to present. It's about reinsurance and reinsurance, maybe some of the people on the call are familiar with what reinsurance is. If you are not, no problem. I'm going to explain everything you need to know. Basically what reinsurers do, they provide coverage to insurance companies and when they provide the contract to cover the risks, they of course need somehow to rate this risk. And there's two different approaches which are traditionally used. One is experience rating, the other exposure rating. And what I'm going to present is a new framework which is integrating the two into one integrated framework basically. And just to what you can see here, what is highlighted, there are three components to it. The first component is a generative model. The second is, I mean, the generative model, he will create artificial or simulated claims. And what we need to do before we can do an analysis, we need somehow to reduce this high dimensional complex objects into some smaller number of what I call reduced random variables. And then we use invasion inference to basically calibrate the model. This is basically what it's supposed to do.

What you see on the bottom, I made a reference to some people in the history.

So the traditional approach where you use experience rating, you're taking the claims from the past, you somehow project this into the future and use this data as a model for the future claims activity.

So that's why I put there Kepler, who basically took the data from Tycho Brahe and then he derived his laws for the

motion of the planets. And then a contrast is Newton, which basically can derive these laws by some much more

fundamental principles. So basically in one case, you model the data and the other case, you model the world. So this is the

difference between the two. Of course, in biology, we would have the Aristotle on the one hand side and then Darwin on the other side and from the statistics, what you see on the left side, this is the father of the classical statistics of Fisher and on the other side, they put the base. So this is where the frame where we are operating. Now to my talk, I will first give some background information and explain why this is relevant. Then I will somehow make a link to active inference, then also point out to some of the challenges which we are facing in this attempt. And then I'll provide with a roadmap for a solution.

And then I will show some examples. And just for your information, I will not show any formulas here.

So whenever you want to look into the formulas, we can go back to the paper, you find all the technical details. So I'll just describe with the help of numbers and some charts or some graphics, what I'm going to

or the basic ideas are. So just if you expect any formulas, you will be disappointed.

Okay, so first, what I've done, I've taken some real numbers from a re-insurance company. And this is Swiss Re.

And from Swiss Re, I've taken not the whole business, but just one segment, which is the PNC reinsurance, which is

property, casualty and specialty lines. Two other segments, which this re-hears is a corporate solution.

So this is then direct insurance provided to large corporations. And there's also life and health.

So what I'm showing here is only PNC reinsurance, so contracts with insurance companies.

And Swiss Re has developed about 20 or 25 years ago, an internal framework for economic valuation of the business. And it has started in 2013 to publish the numbers. So EVM results, which is an economic valuation of the business in contrast to the traditional financial numbers, which have been published.

So EVM results, which is the best way to do. And what is relevant for us here, I'm not going to into details of EVM, but what we have is so-called present values of future cash flows. This is the basis for it.

And then it's the analysis done by underwriting year and not by financial year. So these are the two main of the most important elements of EVM. And what we have is, let's say, the gross premiums and then commissions.

And the commissions are the amount of the premiums, which goes to the intermediaries, the brokers and or goes back to the insurance company in the form of a contribution to the claims or to the internal expenses. So what matters is what they've shown here as net premiums. So this is, it has been increasing for about

from 13 billion in 2013 to roughly 18 billion in 2023. And on the right, you see the mean, and then I use the the mean over the of the net premiums as a base. So this is 100%. And then you can see what happens to these premiums.

So the more the largest part of it is used to pay claims. So this is about 85%. Then you have about 10% is the internal the expenses of the reinsurer of Swiss Re. Then we have some other elements like taxes.

Taxes depend on the on the profit. So it's linked to the bottom line. Then we have capital cost. Capital cost is linked to the risk. But these are relatively small numbers. And then the finance people have introduced some small details which they put on the other. So there is a lot of work going into the small

elements. But what is really matter really matters for the success of the company is what we have here in claims and benefits and how much premium week the company earns to cover these claims. And just as what we see here is the numbers published at time one, so one year after inception. So the 2013 numbers are

published in 2014 based on the information which is available at that point in time. But claims that can take quite some time until they are known and paid and reported. So there is only little information available after one year. So there is some development taking place later. And all the developments of prior years is shown at the very bottom. And we can now see two important things. One is if you go here to the EVM results, you see kind of a cycle. So there is some positive results, then it becomes very negative for a couple of years and then becomes positive again. So this is one thing. This is the nature of the business. And the cycle has two components. One, the premium rates go up and down.

And second, also the terms and conditions of the policies that might be tighter or they might be released. So there's also the claims might change during the cycle. But what is what we see here is the bottom line result. And then there is another element here, the EVM profit, the EVM profit of previous prior years. And here we can see some quite negative numbers. And then here again, this is something which you want to avoid. This is the big surprise, which at the time of underwriting, you put aside a certain amount of the premium for the reserves for the claims. And then it turns out that this is not sufficient. And then you have to find some means or take capital to increase the reserves in later years. And this is basically what reinsurers want to avoid. So of course, what is not shown here is the initial assessment of the underwriters at the time when they look at the business. So they also have an estimate for the claims. This would be the claims at time zero. This is our initial reserves. What we see here is the claims at time one. And what is important to know is, of course, the estimate is quite on a contract or a transaction level is quite high, the uncertainty. But the problem is that underwriting functions, they tend to underestimate the claims during the soft market. So they're too optimistic. And then there are adverse developments. And they overestimate the claims costs during the hard market. So they're maybe too conservative and they might miss some opportunities. And if you just look at this picture here, then you can say there is a potential for improvement in this case of CISRI. I mean, this is my from the outside view, maybe the internal people will say something different. But just looking at these numbers, I would say that adverse developments, so the numbers which you have here, which in the order of one to two billion, this could be improved potentially by about 10% of the premiums. So about, let's say one or two billion per year. So there is quite some potential to reduce this uncertainty. That's one motivation to do this. And the second motivation is, of course, by better risk selection, you could also increase the bottom line results. So this 470 million, the average, but let's say one, which is 3%, maybe for one or two percentage points of the premium volume. So this is the motivation. So we talked about quite substantial numbers in the case of the profit, maybe 100 or 200 million per year, which additional profits could be done, but better risk collection, and we come to adverse development, which is basically the unwanted development in future years, maybe one or two billion. This is my claim. People can prove me wrong, but this is what I think is possible. Now, sorry. Now, if you look at this slide, there is three charts with the temporal development. And what we have done here, I've taken the numbers by sub line of business. So this is also published by Swiss Reels, so I'm not inventing anything. And let's first look on the left hand side. What we see here is different lines of business. So the red one here will be liability. And the top we have here property, and we see they develop quite differently. Now, what is this development which I'm showing here? This chart shows which percentage of the claims is reported after one year. And we see in casualty, it's only about 10%. In property, it's maybe 50%. And then it takes maybe in casualty 10 years, maybe, or even longer until all the claims have been reported. So this is just a reporting pattern, which is lined up with the business dependent. Then reported means that there is an initial, the insurance company, they have notified the reinsurer, there is this and this claim, and they will put up some case reserves and reserves for it. But then it

can take another couple of years until the claims are paid. And this payment pattern is shown here on the right hand side. So you see again here, in the case of liability, it's very slow. So the payments can take 10, 20, 30 years. In the case of short tail or property, this is normally settled within a few years. And then what I have shown in the middle, this is the key topic of today. So one point is the initial estimate of the claims. So this is what underwriting is doing. And the question is what is happening with this estimate down the road. And this is shown here in the middle. And here in the, for the red, which is liability again, you can see here at time one, it's only between depending 70, 80% of the latest numbers.

So there is in time one, the reserves are maybe here 80 or 70%. And then these reserves have to be gradually increased until they reach a stable level. In the case of property or accidentals, at time one, we are higher,

and then it decreases. So here we have kind of a favorable development. So reserves can be released.

And here we have an adverse development, which means you have to constantly increase the reserves in order to be able to pay all the claims. And if we go back here, we can see in the first,

after one year, it's only 10% of the claims have been reported. So whatever reserves we have here is the estimate of the underwriters, there is nothing else, nor their information. So here we can be very confident that the initial estimate of the underwriter was more or less on the same level,

because there is no additional information coming in. So they started 20 to 30 percentage points below the ultimate, which is quite substantial. In the case of short tail lines, it's not so clear,

because you see here about 50% of the claims have already been reported. So it could be that the claims manager, they're quite conservative at the beginning, they reserve more than what is necessary, and then it comes down. Here we don't know, maybe we started higher, it came down, maybe started below, and it came up in the first year. It's not so clear. But overall, we can say some business starts too high, whichever, and then it has a favorable development, some starts just at the right level, and it's flat.

So we have an adequate estimate at the beginning, and in other cases, it's adverse. And this is basically what we want to avoid. Now, there is another complication. And this is that the market is not

stable, the market goes through cycles, as we've seen at the beginning with the numbers of Swiss Re.

And I've tried to show here is if you take each and every contract, you can put the tag on each one.

And say, it either contributes to the economic profit, or it's just adequate to cover the expenses and all the costs. Or maybe some business you even if it is negative, but you want to keep it because

you want to keep the client relationship. So this is then, and the problem is, nobody knows what the true color is. So basically, the underwriters, they take each contract, and they do an assessment, and they say one contract is green, one is amber, one is red, but it could be false positives or false

negatives. This is not known. And of course, if a contract is red, and you say it's green, then you write as much as possible of it, and then you end up having with adverse development. And of course,

you also, then you have the winner's curve. So you're much more likely to get this contract than if having one which is positive, and everyone else also sees it positive. So this is kind of the

broader context in which the underwriting or the market is operating. And if you say that underwriters

are biased, then you can say they are, so there is a true, let's say that the true border between the three buckets, but the underwriters, they believe that the borders may be up here, in the case of a hard market. So they are too conservative, they say some business which actually is already amber, they put it into the

red bucket, and some business which is amber, they would put it, which is green, they would put it into the amber bucket, and vice versa during the soft market. So this is the problematic. And by improving the underwriting, and by avoiding the bias and minimizing or reducing the uncertainty, you can get this dash line here closer to the true border, and here as well. So that's the aim of the whole thing. So this is where I'm making a claim that there is quite some potential for improvement for a reinsurance company. Now there is of course also different approaches in reinsurance. So if you're a big company like Swiss Re, then you have to do these

assessments. In other cases, if you are small, you say I don't do this myself, I save this cost, and I just follow the leaders. Then you just basically can do it for free. And you try to write the same business as the large ones, and then you have it the same at the lower cost ratio. This is also a possible approach.

Or you can say, I don't care over the cycle, the market will correct itself. And I just write and that I'm sure that I can survive the soft market. And then I will earn during the hard market. That's also a possible approach. So we talk here about a company which does a fundamental assessment. And that's why it's necessary to go through this analysis. Okay, now just a few thoughts about how to link this to active inference. And what I've first been thinking about the corporation, so they, the corporate agents, they act very similar to humans. I mean, they're also competing for resources, they have an incomplete knowledge of the external world. They also use the internal representation of the external world, with the help of models and so on. Then we have multiple departments which in our corporation, and they use, of course, they have a different picture of the outside world. And they also have different interests. So they are not always pursuing the same goal. So this is kind of, maybe we feel the same as humans, we are also not always sure. One thought wants to go in one direction or the other direction. So I think this is very similar to corporations. And of course, corporations at the end, they have people there, at least this is still the case. So we have also biological brains driving them. And I think active inference is a very good tool to also understand the behavior of corporate agents in many different levels. But what I'm doing here, I'm just focusing on underwriting or insurance. And

what I'm going to show how generative models can be used in this process. And maybe just if you want to look at a broader role of active inference in economics, then there is one link, which also introduces the term of cognitive economics. So this is where it in a broader picture where it would fit.

So this is where it would fit. Okay. Now, what I'm showing here is what you need to understand about how reinsurance works. So we have the external world where we have objects at risk. This can be buildings, this can be people, this can be airplanes, this can be corporations, which maybe could default and so on. So these are the objects at risk. Then we have the range of the insurance company, which I call here the sedent, which issues policies to cover some risks for these objects. Then there is the external world, where you have the generative process, which creates all kinds of events. And some of these events, like an

earthquake or windstorm or an accident or a defaulting company results in insured claims. Now, the reinsurer doesn't get all this information, which is available on the site of the Sedent company. What the Sedent does, it creates a submission and the submissions, you can put it into three buckets. In one bucket, we have the claims which are reported to the reinsurers. Normally, this is claims above a certain threshold.

The second bucket contains the contract terms. So, of course, you have the, what were the reinsurance conditions in the past, but maybe also some insurance conditions, so insured retentions or some limits and the like. So what is, which act like a filter on the claims.

And the third bucket is the exposure profile, which is basically some statistics about, sometimes also details about the insured qualities and the objects at risk. So in the case of earthquake, you maybe want to know, is my, the building which I'm covering in California on top of a fault, or is it somewhere, I don't know, in the center of Africa, where there is no earthquake risk. And so there it's important to have also some additional information. But what exactly we find in the exposure profile depends very much on the line of business. The gain of claims, this is much more standardized. Basically, what you have is the amount of the claim and the payments, so you can put it into cash flows and some outstanding claims. So this is some, it's much easier to standardize the claims.

So within the exposure, you need specific information depending on the line of business.

Then what reinsurers then have, they, as I mentioned already before, you have two models. One is experience rating. The other is exposure rating. Experience rate takes the claims and the contract terms. The exposure rating takes the exposure and the contract terms. Then there is a quite heavy involvement of the underwriter or the actuary, which is depending who does it. And in both models, you then get some results with some statistics. And since these two results can be quite different, there is some blending process, thank you, place. And then you end up with the projective statistics for the cover year. Then you take into consideration other things like, of course, underwriting consideration, client management, which has some saying, and risk management and finance. So there is now this information is just one element in the decision process. And at the end of the day, you end up with a re-insurance contract for the future. So this information is about the past and the present. You use it to project some statistics and the expectations for the future. This is basically where you have the expected loss cost, which I mentioned before, and which you want to be unbiased and minimize the uncertainty. So this is where the circle is closing. You could also represent it with a Markov blanket. You have some things happening inside of the blanket, the information coming in and some action going out. So this is where active inference is basically visualized.

Okay, now the step to the integrated approach is basically we are looking now into these two parts.

So these two models, the experience rating and exposure rate model. This is the focus area. And what we do here, we have two things. First, we have a generative model. Now this generative model is very similar to the exposure rating model, but the difference is the exposure rating is only applied to the future, to the cover year. While the generative model is basically taking the same model and you apply it to the past. So you don't only apply it for the cover year, this is then comes later, but you first do a calibration of this model. And this calibration is done with the

reported claims. So you use the claims not to project them into the future, but you take the model and project the model to the past. And then you compare the outcome of your model with the actual claims. And then you calibrate the model. And once it's calibrated, you can take this model project into the future and produce all the statistics you need for the future cover year. So this is the basic secret of this presentation. So as I mentioned before, it's a generative model. You have reduced claim statistics and you have Bayesian inference, which you apply to calibrate the model. Maybe you have to go through multiple iterations, but this is basically where the merging is happening. And what I have done in my paper, I've basically looked at this part here, because here, the generative model, you need to come up with very specific model for each business segment. So for each line of business, you have a different data available, you have different models which you are using. So this is where really the expertise of the reinsurer comes into play. So they will have to provide these exposure models or generative models for each segment. But once this is done, then you can basically standardize this part here, the model has to produce some claims in a standard format, you compare them with the report claims. And you then come up with some calibration parameters, which you feed back into your model. And once this is calibrated, you can then apply to the future and do everything you need to do on this right hand side. So that's basically the magic here. And we will see it's much, you need much less input from the underwriter, because here in the past, when you need to, when you do experience right, you have to project these claims to the future. So there's a lot of assumptions going into this process, because the, if you go to the past, the portfolio was different, maybe the claims that in the past are no longer representative. So there's many, many assumptions which come from the underwriter, and this can be avoided in this approach here. Okay, now, so far, it sounds relatively easy. But of course, if it was easy, maybe it would have been done. So there are some challenges which you need to resolve. And to understand this, what I've done here, I'm just taking a few simulated claims. And where we have here is the time axis. In the vertical, we have the incurred amount. So the estimate of what the ultimate is going to be. And in the y axis, I have the paid amount. And let's assume here at the end, it's settled. So you can see the estimate. So first, it goes up, and then it comes down. And then there is some adjustments. And then it goes up again. So you can see this in this curve here, the paid, first, there is nothing paid, and then it goes up. And then eventually, we have the final payment. So this is one example. But what can also happen is that at the beginning, you know nothing about the claim. So the claim is reported with some delay. So the accident date is here in 2016, but the reporting date is 2020. So in this time, you have to put up so-called IBNRs, so incurred but not reported reserves, for all the claims which will be reported later. You have also seen this at the beginning, the case of CISRI, the reporting pattern was the chart on the left. So this is the nature of the business. Now, if I take one claim, so the whole graph here is one claim. And you can either represent it with a graph in the three dimensional space, as I did here, or you can take each payment and each incurred adjustment and say that's just one element, and then you put it into a high dimensional space. So you can say that each claim is an element in a high dimensional sample space. And the dimension here is maybe something between 10 in the best case,

or can it be up to 100. So it's very high. And it's just a high dimensional space where it's very, very thinly populated. Imagine you get about maybe 100 or 1000 claims per submission. And if you have a claim with 100 or 50 or even 20 dimensions, you're completely lost. There's nothing you can do. So what is done here, and this is maybe the only formal of the same sample symbols I'm using here, I say the sample space is high dimensional. And what we do, we decompose it in M one dimensional spaces. That's basically the whole idea behind. So we need to find some variables which are more or less independent. And we project these claims to these subspaces, and then we can solve the problem in a one dimensional space, a couple of times. But it's identity with a one dimensional problem, which is much easier, of course, because then we have 100 to 1000 claims in a one dimensional space. And this is easy to handle. So this is the second part of the reduction. And I will show you how this is done. But just to understand that this step is necessary, else you there's nothing you can do. Now, if I go here, how can we solve this? So this is again, another representation of the standard approach. So what you have down here, you have claims which are in this case 10 years old. So I use minus for relative to the underwriting year. So it's underwriting year minus 10. Here, these claims have been you have a reporting pattern of 10 years. So you know, the first 10 years of the development, but for the most recent year, so which is minus one, you have only one year of development. So this is now if you try to do traditional experience rating, there's two things you need to do. You need to project all the past to the future. So this is one step. Yes, you need to trend the claims and say on an as if basis, a claim which happened in the year minus 10, what would it be if it would happen in the year plus one, which is basically my cover period. And then what you see here is we get this kind of triangle. So some claims have 10 years of development, but the most recent ones, they have only one. Now the problem is that these recent claims are maybe representative for the portfolio, but they have very little development. The old claims, you have a lot of development, but they are no longer representative. And now what you need to do is you need to take this triangle and project it to the end and see what is the end of this thing. So there's a lot of, as I said, a lot of underwriting assumptions going into these two, these fundamental steps. And this is the experience based approach. In the case of exposure rating, where you take the information, which is available at normally the minus one. So the market experience, these are the market models. You have the exposure profiles, insurance conditions, you project everything to the cover period and you apply your model and you get an exposure based result. And then you take the two, blend them and that's then the result. I can take the same now and represent what is it in the integrated approach. So again, the three steps. So first I now use my exposure model, not only for the future, this is then the last step, this is what I will have to do. But before I do this, I'm calibrating my model. So I apply the model to the past. So I take all the market experience which I had in the past, the profiles from the past, the models which can change over time and the

insurance conditions. So I have a one model for each year. Some of the information might be the same, but basically I can now run this model for the year minus 10 and I can create not only one but 100 or 1000 or as many as I want simulated sets with claims. And I do this for the entire period. So I have here now a large number of sets. I could say it's k different sets and I have one observed set. And now I can say the general statement is my model is calibrated if this one observed set of claims is not an outlier in the set of all simulated claims.

So if this is one element or basically one member of this family here, then I can say yes, I cannot reject my model because it is compatible with the data. And once this is the case, then I can project it to the future and run the model. Now the problem is now doing a comparison on this level. So to compare all the claims which are in here with all the claims that are in here, we are operating in this high dimensional space and there is basically no way to do it. So what we do in a first is we reduce the claims. So this is happening here. And we represent the simulated claims in the same way as we represent the observed claims. So we use the same statistics.

So from the latest year, I could, of course, I could simulate it till the end, but I'm only taking the first year. So the system knows behind what's happening to the end, but for the analysis, I'm only taking the first year. And if I go back 10 years, I take the first 10 years of the simulation. So basically, I have exactly the same representation of the claims in the simulated sets and in the observed set. Now I project it, as I said before, to these one dimensional spaces and then I can do the analysis and I can come up with calibration parameters. This is how this basically works. Now we can have now a look at how do we reduce the claims. And I have shown here two very simplified developments. One is the reported pattern and the other, or the incurred, and then the paid amount. So the, in the case of the incurred, there's again, it takes a couple of years until the, or some time until the claim is reported. So I've here a reported lag. And then the claim, then the first estimate of the incurred. And then here there's an adjustment and here again, and then it got, this continues into the future until the claim is closed. But of course, this information I don't have. So what I'm taking here is the latest estimate for the ultimate. This is what I call it, I , for the incurred star for the case simulation, for instance. On the right hand side, I have then the payment patterns. And what is important to make sure that the payments are always smaller than the incurred because usually as there will be, this is one of the boundary conditions in the, in this information, but we don't really care about the individual numbers. What we do is we take here an average legs. It's basically the area which I have here above the curve. So this gray area here, if I divide it by the amount, then I get the incurred leg. This is this part here. This is one of these reduced numbers. So I'm not looking at the details. I'm just taking the average leg, which I have here. So this is one figure I'm taking. And then another statistic I can use is the number of adjustments, which I have seen. This is one, two, three. That's also one number I'm using to calibrate. So to say how many adjustments is an insurance

company making for the claims? Is it larger? Is it small? So these are the reduced statistics here. So the reporting lag, the amount of course, the incurred lag and the number of adjustments. And on the paid side, I can take very similarly the paid amount or the ratio which is paid. I can also take a paid lag and also the number of payments. So these are some of these reduced random variables, which I'm using for calibration. So I'm simplifying and these numbers are considered to be largely, they're not completely independent, but they are largely independent. Now here is another representation of the process. So we again, we have now at the bottom, the events which are leading to insured claims. We have the set with claims and the part which is observed. So this is more or less what the seeding company is reporting to the insurer. And then we have a set of reduced claims. On the other side, we have now the generative model, which is creating the K sets with simulated claims. This can be very detailed. There may be in a first step, you align these claims representation with the observed claims and then in the next step, we reduce them. And then we take this information here to come up with a conditional distribution functions. Then you take the observations and you apply Bayes and you end up with the calibration parameters, which you can feed back into the model. So this is another representation of this process. And once the model is calibrated, then of course, you can go and apply to the future and do everything you need to be able to rate your business.

Okay, now let's look at some results.

First, what I've done here is some very simple. I've used, in this case, not actual claims, but I've used a simulator for the generative process. And what I've done here, it's a log normal process and I assume

there is some trend. So I go back the years and I have the true probability distribution for each year, which is represented in the vertical axis. You see here the probability. And I've drawn one random number for each year, which is shown here with these small diamonds. And what we also have here is then the projection to the future. So assuming this trend continues, we have then the, this is then the true, in this case, for the generative process, which is simulated, but this is the true distribution in the future. Now, here I have my generative model. And you can see my initial model is flat. So there is no trend in there. And what I can do in a simple approach, simply say, okay, let me try to adjust the level. So I'm just moving this model up and down and try to align it with the data. This is done down here. So here we have then the, again, a flat model, but you can see at the beginning, it's too high because the observations are on the lower side. And at the end, it's too high because the observations are on the higher side. So actually, there seems to be some trend in the data. And which, of course, it reflects the trend in the underlying model, which is unknown. Now, if I can also do is to say, okay, I am trend, I'm, I'm calibrating the level and the trend. And this is what I've done here in this third picture here. Now I have, you can see a model which is increasing over time. So this is the generative model. And you can see now it's well aligned with the data. And if I compare now just the, the first two moments, so the mean and the standard deviation, then here I'm quite close to the true hidden model. So this is basically unknown, but this is what I've put behind the scenes. The true

model is 83 with a standard deviation of 8.3. And here I have 88 with a standard deviation of 8.8. So it comes pretty close here. My initial model was 60 with a sigma of 6. If I calibrate only the level, I end up with 72 and the standard of a sigma of 72, or 7.2. So you can see if I use this approach, then I can get very close to the hidden true model. Here is a second example, which is now a compound model. So I have a simulating the number, I take a negative binomial distribution to simulate the number of claims and then the log normal distribution to simulate the severity of the claims. And this is again shown here. And the same approach, you have the, just calibrate the level and you calibrate the level and the trend and you get pretty close to the initial model. So just a different process behind, but the principle is the same.

So here I have another model approach where I show you a little bit more in detail what's actually happening. Here I have again a negative binomial distribution for the generative process. Now here I've shown the probability mass function. So the probabilities for each outcome. And what you show, see here in the second part is then a log normal fit to this distribution. And I could, I could do the analysis with the discrete distribution, which is possible, but it could be computation a little bit more challenging, but it's doable. But what we can see, as long as we, we want to assume that the observations are in

the high density region of the model. So we can easily replace it by a fictive distribution, which is a much easier to handle because here I can use a, um, um, um, oh, can I find the term?

So the whole basin analysis is basically much easier if you take a log normal distribution than compared to a negative binomial. Now here again, down the same. Now here, what I have the probability distribution is not

the model behind it directly, but the simulated, uh, models where I have the take my generative model. I'm simulating

a large number of claims, and then I take the empirical distribution as a basis for my analysis.

And here again, I fit a log normal distribution. So we're just fitting the first two moments.

And then I do the analysis with this, um, fitted distributions. What I've also shown here is then I've aggregated the whole 10 year or 12 years here into the, uh, into the mean and shown the distribution of the mean. Um, and I can use this distribution to calibrate the level if I want. Yeah. So the level calibration I can do by using the annual data, but I can say I'm just looking at this overall distribution and I calibrate it. So this is when you see here, then you see the observation and calibrated observation is more or less in the center of the distribution, but still, if I look then at each and every year, then I still have this, uh, trend, which is missing. So if I am calibrating level and trend, then you can see here, I come up with a model, which is very close to the initial or to the true hidden generative process. So this is, um, the, how it works. So you take your generative model to create the, um, simulated distributions.

You fit, uh, discrete, uh, sorry, um, a continuous distribution to it. And then you apply base by using your priors and to find out, you can find out the calibration parameters. Now here, what I've now put everything into, um, uh, one chart to say, uh, at the end, what we have is we have

seen, we have different, um, reduced random variables. So here's the number of reported claims. Then we have the reported lag and the incurred amount, the incurred lag, the number of incurred adjustments. Then for the paid, we have the paid lag and the number of payments. And what I've shown on the right is some, then additional, uh, have a strongly dependent variables, which are, we can use as a, to observe, to check if the model is okay. But in this case, I've only calibrated this green area here and here, I've just shown one, uh, hand where I can change the level and another one where I can change the trend and maybe I'll run multiple iterations. So in this case, I think I run it five times to basically, uh, be able to deal with the dependencies. And what I've shown now is the, I don't know if you can see here, this is now the, the distribution, which I'm using in as a, as a prior. And then behind, this is in light gray, you can see the true distribution.

So you see here, there is an offset. And after calibration, you can see these two distributions are basically overlapping, which means there is no gap anymore. And this is, here is another big gap here. It's already at the beginning. Okay. Um, and so you can see the calibrated model is basically very close or the, the deviation like here is smaller than one standard deviation. So this is basically as good as you can get. So this is the, um, how the whole framework works. So we apply it to multiple random variables. So the ones I'm introduced before. So the, which is shown here, you can have some additional random variables, which you just monitor, and then you run your, your calibration model until you are satisfied with the result. And then you apply the model to the future.

Now, just to summarize, I've looked here, this also can find in the paper, the comparisons of the integrated framework, what are the pluses and the compared to the standard approach. And you can see it is, I mean, you use all information in a coherent way. You don't waste any information in the standard framework. There's a lot of information you cannot use, and it's not aligned. So it's, uh, this approach here is superior in all regards, but there is, of course, it's not for free. The price you have to pay is the, the computational effort and also the effort to set up the whole system, because you need to be able to run it to the past. Of course, this effort is much higher, but you have the benefit that maybe you can, so the substantial reduce the uncertainty and the bias in your estimates. So this was the motivation which we addressed at the beginning. Okay, here's just the summary. So basically, we have this, uh, the process and the traditional model and over here the integrated approach. And it's a kind of a paradigm shift, which requires that you have as a new mindset to approach it, because the actuaries, especially in certain, um, in the Anglo-Saxon region, they have well-defined processes, how they are supposed to rate the, and assess the re-interviews basis. This is, you will not find in any of these, uh, um, documents which are used. So it's a completely different approach and this will, the market will probably, uh, is in some areas maybe not open to it. So you need, you need a different mindset, you need the different skills. So I think once you want to introduce something like this, you need to run this in parallel to traditional approach until you realize that it has a benefit. And maybe it's a similar process when, as in Swiss Re introduced the EVM framework, it took about more than 10 years until they went public with it. So these things can take quite some time, but maybe there is a new technology which is around in artificial intelligence. Maybe there is some way to

to shorten this transition period. Okay. This is my part. Um, so thank you for the attention and I think we're open for the discussion and questions.

Thank you for the great presentation. Anyone in the live chat can write a question. Otherwise, Roland, it would be great for you to say hello, give any remarks and a first question. And then Scott, thank you. Sure. So first of all, congratulations, Stefan. So, uh, Stefan used to be my boss at Swiss Re and, uh, he's one of the OGs who came up with the, uh, with all the pricing frameworks that are still in place today. And, um, uh, so I'm also, I was super happy when I met him again and he, he showed me what he's been up to

here. And I think that's really where to take it because when I look back in the last 20 years or so, there were like almost no developments in this area. And, uh, I think, uh, nowadays, uh, with all the AI happening, it's, it's really time to do something like this, but now comes my biggest question to you, Stefan, do you think that the world is ready for this? Do you think that anyone will look at this, that there is any, any form of acceptance for this? You see, I'm very, I mentioned before, I'm very skeptical because we have, um, the standardized approaches and I have already tried to also to approach some, I mean, I left Swiss Re about 10 years ago and I was doing something completely different to me. And this then came up somehow as a side product of other activities. And then I remembered I had this problem in the back of my mind, which I wanted to already address at that time. I presented some ideas at the time and nobody was open for it. And if I, by talking to some of the former colleagues, I have the impression they are not ready. That's why I mentioned here, it requires a new mindset and maybe a new mindset requires new people. And I don't know what has to happen to the industry. Maybe there are some smaller companies which say, oh, great, this is an opportunity for us. So let's jump on this. But I think the old dinosaurs, so far, I don't see any willingness to change their approach.

Okay, Scott, if you'd like to give any initial remarks, I know you wrote a novel in the chat, but if you'd like to give any remarks and maybe a question somehow concisely. Daniel knows that when I write a lot in the chat, it's an excitement. So Stefan, first of all, thank you for fantastic presentation. And it's a, I feel like first saying, where have you been all my life? Because you've given voice to a lot of the things that felt, ever since I've heard about active inference, felt like they needed to be thought about in risk. And so I was a corporate and tax attorney for 30 years. And so I see these things. And so my, I think my main question is, and what is one of my latter questions in the chat? I'm always looking for ways to introduce new things where you don't have to rip and replace. Because people who, I remember talking to the head of security of AT&T one time, and he said, why would we do anything different next Tuesday than we did last Tuesday when we made a billion dollars? So the, so how do, how do we get people to want to do something? And what I wonder is as a immediate matter, could the generative model, the application in your slide right here under the Beijing approach of projected generative model to the past, could that be included as a calibration in effect, but on the existing projection of claims history to the future, that analysis? Is there a way to make the thing that we're aspiring to migrate towards a part of an existing part of the analysis,

so that it's seen as being additional, but not requiring migration to the new analysis entirely? Is there a way to bolt this on so that people can start getting used to it, seeing its value, in the context of the analysis that's already being done, or are they so fundamentally different that the new approach doesn't lend itself to being offered as an add-on in existing approaches? What are your thoughts on that? Yeah, very good question, but if you look at these two slides, just over here, yes, and over here, both approaches use more or less the same information. Of course you might say, okay, here maybe you have a different model than over here, but it could also be the same model. So you can, I think it's necessary that in the transition phase, you apply this new concept in parallel to the existing methodology. So the people can also get the feeling, what is the difference, and then it will take some time, I don't know how many years, until you see the results, because you will not see the results immediately. It will take some time until you see that, yes, down the road, we have really much less adverse development in the claims, and this you need, maybe you need about five years of experience, until you see the results really materialize. And this is probably also the time it takes for the people to familiarize with the methodology. So I think we want to implement something like this, you need to do it step by step, maybe you don't do it at the beginning for all lines of business, you maybe confine it to a few lines of business. And then over time, you can expand to some other. And of course, people need to get used to it. And once so you run the old framework in parallel, and then you can compare and it's time you get the sense for it. And once you feel comfortable, then you can basically turn off the old part.

Well, thanks again for all, just give a short note. This slide really captures the difference here, going from extrapolating descriptive statistics on observables into making a generative world model that's based upon or conditioned or updated from observables, but also has latent states. It's we don't remember the time series of visual input and then extrapolate. We update based upon the observations and then use that to do the now casting and the forecasting. So that has all kinds of beneficial properties. The reduced representation and the ability to explore direct counterfactuals, generate synthetic data. There's just a lot of things that are purely time series correlation based approach.

It doesn't really get at. The question I wanted to throw out to all of you was, as a non-reinsurer, coming from a biology perspective, what are biological analogies to the reinsurers or the underwriters role? Like, could it be considered to an epistemic foraging portfolio, like research projects, could it relate to individuals and their weighting in a group, like risk profiles of different foraging strategies within a group? Or how does it happen? Because that idea of being too or too little cautious in a changing context with the hard and the soft market, there's a possibility of being over or under risk averse in either setting. But you don't know the true risk setting.

Maybe let me try to first give a, provide just a short thought. If I look at biology, then I think biology, you find it over here. And I don't have difficulties to come up with an analogy in biology, which, which correspond to this approach here. But maybe someone else has a better idea. But maybe just one analogy with something I did here is it's kind of intentional I put here, you see one part of the

information into red, one into green and one into blue. So you could say this corresponds to the cones you have in your eyes. So basically, you also have three receptors for different information, which of course, it's just now it's not really one to one comparison. But this is maybe what I tried to do. So you have the receptor, you combine this, this thing, you come up with the internal view of the external world, and then you act on it. So this is the analogy as far as I can see it, but maybe someone else has a better example in mind.

I mean, at the, at the broadest level, sorry, Roland, did you have something?

Roland Barthes- No, just go ahead, David.

David Barthes- I was gonna say, at the broadest level, I put it in the chat.

I mean, it feels like so I used to work in securitizations. So you take credit card portfolios or car finance portfolios and put them into packages and sell them off. And that used to occur to me that what you're doing is trying to find a bigger sucker for the risk, right, you package it up, you make it look good. And that was the 2008 financial break, was packages of mortgages. They were crappy mortgages because the banks were enthusiastic about selling off the mortgages. So they made the mortgages so they were appropriate for sale in these packages. But they weren't necessarily good risks. So the risk just got iterated up. And what I'm wondering, you know, you think about a biology, what do you what are the name of the organisms that eat dead decaying matter, like leaf litter and stuff like that?

David Barthes- Whatever. They, I mean, that death, it's everything's recycled. And risk is recycled continuously. That's what biology is. So you have this big negentropic urge that negentropic beings have, whether they're physical or not. And they're all dealing and everyone loses, ultimately, their own individual battle on negentropy. All organisms die. But then they're recycled by other organisms. And so they're copper fights, you know, that eat poop, you know, things like that. I mean, there's all sorts of recycling that happens. And so it feels like, from a reinsurance perspective, you have abstractions of risk. What you're with all insurance, you're creating instruments to create a, a, an organism, a symbiote, a sim infogenesis organism that can cause you to de-risk in ways you can't do alone, period. So it's like a eukaryote, right? You're adding, people could just not do reinsurance and not do insurance and self-insure, and then have to pay the full price of their house burning down or whatever.

But we have these abstractions where we kind of develop these eukaryotic things and say, hey, let's do this together. It's a good idea, because then I just pay a premium, I get a house back, and then the reinsurers say, hey, let's do this together. We got all this different flood kind of risk or whatever, and we can do an evaluation at a metadata level. It feels a lot like exactly what happens in biology without memos and voting and all that. And so the power of what you're talking about in revealing Stefan feels like this is how stuff works if we don't get in our own way, politically and with false models from a history of economics, etc. And if we really allow for the information to flow among the entities at risk, doesn't mean that every organism survives forever.

It's not about perpetual motion or perpetual life, but rather there's a balance that happens naturally. So that feels like as an immediate matter, there's an ability to sell this to the quants on Wall Street and say, hey, your model is slightly out of whack with the biophysical physics reality, the mathematical

physical reality. And so, hey, you're an arbitrageur. If you want to make money on the difference between reality and your model, here it is. You don't have to do it to do what you're doing.

Prof. David P. So it feels like it could be pretty attractive in the right kind of audience for people who have to deal with risk professionally that they can't turn away. Once you say something to them, if it has value and shows value, they won't be able to say, oh, no, we're just going to do it our old way and their clients won't tolerate that. Anyway, sorry, I got a little excited there.

Prof. David P. Yeah, but you go in the absolutely correct direction.

Prof. David P. I mean, I've had to calibrate a lot of risk models in my life. And I mean, when you look at the definition of risk, it's what happens above the expected. So normally you subtract the expected. And normally you have to answer questions like what is the worst that can happen within a hundred years? And hardly any one of us has ever lived a hundred years to actually tell, right?

Prof. David P. And so actually what you learn very quickly when you're a risk professional is that you have to break down the question to something that fits the experience horizon of a living being. And so typically when I was calibrating risk models, I saw, okay, this guy has been in his profession for 10 years. So I was asking questions around what was the worst thing that happened

in 10 years? And then given a model, you try to extrapolate. So what's the worst that can happen in a hundred years? Right. And in my opinion, this approach here is just doing it much more systematically. You know, so we always had a bit of a tendency to do this, but it was more heuristics and this kind of gives it a more scientific way of looking at the problem.

Prof. David P. That's awesome. I'll just make one quick.

Prof. David P. Sorry.

Prof. David P. Oh, go ahead, Stefan.

Prof. David P. Yes.

Prof. David P. I mean, you mentioned, Scott mentioned the financial market. You know, when I joined the insurance industry, this was many years ago, the first thing which happened was the collapse of the so-called LMX market.

Prof. David P. And this was the London excessive loss market. And what they did there is they just handed the risks from one to the next and it went around the circle many, many times. And then there were claims and the claims were also circled around. So the same claim you had already, suddenly you get it by the back door again and you put it on top of it. And this happened until someone ran out of cover.

Prof. David P. And then there were some big losses then happening, some mental events, some other things. I think there was a Piper Alpha, it was windstorms in Europe, there were hurricanes and so on. And suddenly this whole thing collapsed. So it was working before because the circling these things around took very long, as long as it was with paper. But then they started to introduce the new technology. So there was a time of computers, the first computer show. So the whole thing went much faster. And the whole thing collapsed. And so the industry had to learn already in the past. And I don't know what has to happen for the industry to maybe go through a new leap and to start to implement new

approaches. Because as long as there is nothing massive impacting, as you also have mentioned before, then maybe people are just saying, okay, why should I change anything? Everything is okay.

Prof. David P. And if I can, so I put in the chat, that's a direct relationship, but what you're talking about there, to climate change. And let me tell you what I think is going on here. And I think it's pretty wild. So we talk about abstractions. We have abstractions upon abstractions. You got futures markets and derivatives, and they're trading upon trading over trading, and there's all intangible. Now there's a connection ultimately to some physicality, some commodity or some phenomenon. But there are abstractions that are layered on abstractions, and that's fine. But the challenge is we're trying to de-risk and create negentropy. We're dissipating information differentials in those abstract areas, and that requires energy. AI requires coal burning. And so the energy required to maintain the energy generation in physical space, coal, oil, etc., required, and the agriculture required to feed the people, all that stuff that's required in physical space, to keep taking these exponentially increasing levels of abstraction in intangible risk space. Because the financial markets are one set of instantiations, but reinsurance is an abstraction on top of an abstraction. The insurance didn't exist in all forms forever, and it keeps getting more abstracted. Forward contracts are a form of insurance. When you're hedging, you're doing it for insurance. Speculation in markets like that create liquidity, but also make it noisy in terms of what the risk profile is, right? So it feels like as we're de-risking in abstract space, information space, we have, it requires energy that we have to burn coal for in physical space. And so that's, it's an insidious problem that we continue to try to do that. I have to go get something at the door, but I wanted to make that point about that relationship between climate change and the information space.

Thanks. One key point, Roland, that you mentioned was fitting the experience horizon to the living beings. In some ways, that's the cognitive approach, which is fit it to the cognitive slice, like the sense-making and decision selection in the moment.

And then there's the consequences for the kinds of AI or synthetic intelligence models that are made. For example, downloading all of the internet's plain text data, that's more experienced than one human life learning. And then using correlative aspects of that large data set to then carry forth those correlations forward. That's like the LLM framework. And then the question is, how and where, alongside or differently, does a small generative model or some other kind of nested or structured generative model come into play that is able to do better or differently than just recapitulating correlations plus, plus, plus?

And how that actually plays out. And then I think there was a great comments about how does it really get adopted or how does it be shown to make a difference?

And it can be used in retrospective meta-analysis, possibly with the kinds of data that you've shown today. So having open source and clear methods for doing the retrospective and saying, not saying that doing anything differently would have been better,

but here's how this pattern of outcomes and decision-making differed from this.

So give a meta-analytic framework for longer past time series, do backcasting kind of like in stock trading, and then open up that space of, well, people are familiar with stock trading and investing based upon the mean value.

And this moves into also thinking about the variance profiles.

I find this extremely interesting as an observation.

So when I used to work also at Swiss Re together with Stefan and I used to head an initiative in casualty.

And you see in property, things are somewhat objective because you have earthquakes and you have windstorms

and they have kind of a physical nature and you can study how the earth works and you understand it.

Something like liability is much more man-made.

And so essentially it's all about things that are not supposed to happen.

So you always have like an infinite configuration space and you cannot model it.

And already at the time, it was like 20 years ago, but I said, we need to have like monitors

to systematically scan the media, what's being published, what's the awareness,

because with a certain likelihood, the lawyers are going to pick up what's in the media

and they are going to do lawsuits about it, which then result in liability claims.

And there I see exactly the applicability of LLMs and all this stuff,

because what they can do extremely well is put things in a correlation with each other.

Right.

And so I think especially for modeling casualty business, I see a huge potential in what you are

saying. So just, you know, putting LLMs on top and finding out any form of dependence

between what's being published and lawsuits, et cetera.

So Stefan, where are you going to go with this work or what are your directions?

Okay.

A good question.

But, you know, as I think I briefly mentioned,

this, see, I have to maybe just make a short comment about my career.

So I have a degree in physics, but then after my PhD, I then went into reinsurance industry and I

left the reinsurance industry about 10 years ago. And then I was doing completely different things.

But this topic here was kind of something which was brewing in the back of my mind.

So by exploring other fields and maybe I have to say, this is also thanks to my wife, which she exposes me to all kinds of topics. Suddenly I got familiar with different approaches, how to solve problems.

And then suddenly this, I saw a way how to solve this issue here. And my intention was simply to bring it to paper, to write it down, to publish it and get it out. So basically saying, okay, then for me, this is done.

And then Roland came into the picture and he is now coming up with some ideas that maybe we can do something with it. So I don't know exactly what's going to happen.

I, as I also mentioned, I approached some colleagues from former colleagues, but there was no big interest.

And I've also seen in, when I published the paper, the peer reviewers, they say, yes, it's interesting, but for them, it was not something they say, yes, we need to move in this direction.

It was simply asking what is then the benefit compared to the older approaches.

So, so far, I don't see really the, I haven't encountered anyone which is open to it. And I didn't have any plans because I have other ideas, which I want to pursue and retired and I have other projects.

But depending what is happening, if people approach me and say, yes, how can we somehow implement this framework, then I'm open to be involved there. But I have no specific plans to do anything with it anymore.

Maybe some reinsurers listen to this. So I think obviously it's, it's not very nice to do it, to do a dry exercise with fake data.

So we should have real data and we should have a real interest behind it. But maybe if some, some reinsurers are listening to this podcast,

please contact us and let's see that we can collaborate in this, that we can do something in this.

Yeah, there's, there's a lot of things to say. There's this

laid out calculation, and the kinds of Bayesian analyses that are described with this

kind of baseline structure. And then as you alluded to, that's even before getting into

multi-agent cognitive economics, thinking more about the decision making, the sense making of different entities. This is just taking in a very specific approach to structuring the data and the input.

And it's an interesting direction to see what, what tools and how general does the toolkit have to be.

Like when you say external data comes in, that could range from nothing to all the stock data in the world.

So how do you even exactly go about that to make it useful without just saying, well, it could be tiny,

or it could be all the data analysis in the world. And then again, how does that connect to doing

similarly or better or different than the kinds of models already in play? And then the decisions coming out of those.

I mean, it's funny because, oh, go ahead, Stefan.

Stefan Kuykiewicz

I'm sorry. I think you make here, bring an interesting thought into the picture,

because what I see, you can see, I focus on a very tiny part of the whole decision process.

But of course, you can expand now this market blanket approach and make, say, okay, at the end,

we have within an reinsurer company, there is many different buckets, which we could try to understand from this, from the perspective of this new framework, and see how is the, the corporate

decision processes, what are they happening? How do different place, part of parts of the company

interact? How can the company interact with when there's other information coming from the outside,

there? I mean, you have other stakeholders, I mean, I just looked here just as the insurance clients,

but there, of course, there is the financial part, you have the maybe some exchanging risk, maybe you

have the stock, the shareholders and so on. So there is many different facets. And you can also see

them yourself as part of a bigger economic framework and see what is your place in there. So at the end of

the day, I think if you want to have kind of digital earth, you could try to expand this kind of thinking

into some much bigger framework and try to understand the, but I don't know if anyone has the capability to

do it, because it sounds pretty complex, but maybe with the help of artificial intelligence, we might get there.

Yeah, this is another thought, because every time we have these tech revolutions, like now with the AI revolution, it also shifts industry players, right? So if we look at the invention of the smartphone before phone companies were called like Ericsson or Nokia, and afterwards they were called Apple and Google. And so I think there are always these shifts when tech revolutions come. And I think this AI revolution together with such modeling, it could actually then suddenly become attractive for big tech to start doing reinsurance. Because what you need is you need capital, they have capital, and you need skills. And they are actually quite good at acquiring data and analyzing data. So maybe it's exactly this type of model that makes it accessible for big tech now to enter the domain.

Reinsurance on digital information.

Interesting area.

Yeah, I mean, I'm going to look at the scope of reinsurance. It's very, very simple. It's very saying we need someone who absorbs is the last resort for risk, you need to be able to assess this risk. And in the case, if something happens, you need to be able to pay. And I mean, this is the traditional approach, but this could be easily replaced as Roland said by other players, which step into the market. And since it's not largely regulated, this entry point should be relatively easy, if you have the technology and if you have also the financial strengths behind and the, of course, the skills, this is the important part, then you can basically replace all the reinsurance industry.

And it's not complete fiction. If you look at Alibaba, this big Chinese Amazon, so Alibaba started selling insurance. So it's not completely off because they have a platform. So I could picture, you know, Amazon or Facebook to start selling insurance. It's not completely science fiction.

Okay. Very interesting. Roland and Scott, if you have any last comments, then Stefan, you can have the last word. Okay. All good, Scott. Okay. What is your, what's your last word, Stefan?

Okay. Well, I didn't think about having a last word, but I mean, for me, it was very interesting.

I was having this discussion. It was for me as a new experience to be in a podcast and to present in this way. I mean, of course I was presenting many, uh, traditionally, and it's now I'm entering it my late days into a new, uh, platform, which was quite, uh, an interesting experience. So I enjoyed it. I haven't this conversation and let's see what's happening out of it. Great. All right. Thanks again, all till next time. Okay. Thank you.

Thank you.

Thank you.

Thank you.